

Supplementary Table 4. Contingency table of functional-classification evidence across all truth sets and class-assignment methods.

Likelihood ratios and Tavtigian OddsPath values, each with a log-Wald 95% confidence interval and the corresponding ACMG evidence-strength tier, for every functional-classification method, control set, and functional class. Class definitions and assignment methods are given in the footnotes.

Assignment method	Functional class	Class definition	Control set	Path. (in/N)	Ben. (in/N)	LR	95% CI	Evidence	OddsPath	OddsPath 95% CI	OddsPath evid.
GMM score cutoffs	Abnormal	score $\leq$ -0.0286	ClinVar Feb 2026 (all variants)	184/185	25/1171	46.59	(31.61–68.67)	PS3 Strong	44.59	(30.48–65.23)	PS3 Strong
GMM score cutoffs	Normal	score $\geq$ -0.0197	ClinVar Feb 2026 (all variants)	1/185	1133/1171	0.0056	(0.0008–0.0395)	BS3 Strong	0.0111	(0.0028–0.0441)	BS3 Strong
GMM score cutoffs	Abnormal	score $\leq$ -0.0286	ClinVar Feb 2026 (missense only)	21/21	0/28	56.68*	(3.63–885)	PS3 Strong	27.68	(4.03–190)	PS3 Strong
GMM score cutoffs	Normal	score $\geq$ -0.0197	ClinVar Feb 2026 (missense only)	0/21	28/28	0.0231*	(0.0015–0.3584)	BS3 Strong	0.0471	(0.0069–0.3199)	BS3 Strong
GMM score cutoffs	Abnormal	score $\leq$ -0.0286	ClinVar 2025 (all variants)	245/246	26/1065	40.80	(27.91–59.64)	PS3 Strong	39.16	(26.98–56.84)	PS3 Strong
GMM score cutoffs	Normal	score $\geq$ -0.0197	ClinVar 2025 (all variants)	1/246	1025/1065	0.0042	(0.0006–0.0299)	BS3 Strong	0.0084	(0.0021–0.0335)	BS3 Strong
GMM score cutoffs	Abnormal	score $\leq$ -0.0286	ClinVar 2025 (missense only)	13/13	0/11	23.14*	(1.53–350)	PS3 Strong	11.14	(1.70–73.18)	PS3 Moderate
GMM score cutoffs	Normal	score $\geq$ -0.0197	ClinVar 2025 (missense only)	0/13	11/11	0.0373*	(0.0024–0.5682)	BS3 Strong	0.0779	(0.0117–0.5191)	BS3 Moderate
Waters: SGE FDR + directionality	Abnormal	depleted	ClinVar Feb 2026 (all variants)	184/185	73/1171	15.95	(12.77–19.93)	PS3 Moderate	15.67	(12.56–19.54)	PS3 Moderate

Assignment method	Functional class	Class definition	Control set	Path. (in/N)	Ben. (in/N)	LR	95% CI	Evidence	OddsPath	OddsPath 95% CI	OddsPath evid.
Waters: SGE FDR + directionality	Normal	unchanged + enriched	ClinVar Feb 2026 (all variants)	1/185	1098/1171	0.0058	(0.0008–0.0407)	BS3 Strong	0.0115	(0.0029–0.0456)	BS3 Strong
Waters: SGE FDR + directionality	Abnormal	depleted	ClinVar Feb 2026 (missense only)	21/21	1/28	28.00	(4.09–192)	PS3 Strong	13.84	(3.62–52.88)	PS3 Moderate
Waters: SGE FDR + directionality	Normal	unchanged + enriched	ClinVar Feb 2026 (missense only)	0/21	27/28	0.0240*	(0.0015–0.3717)	BS3 Strong	0.0488	(0.0072–0.3322)	BS3 Strong
Tejura: SGE FDR + directionality	Abnormal	depleted	ClinVar 2025 (all variants)	245/246	70/1065	15.15	(12.08–19.01)	PS3 Moderate	14.89	(11.89–18.65)	PS3 Moderate
Tejura: SGE FDR + directionality	Normal	unchanged	ClinVar 2025 (all variants)	1/246	910/1065	0.0048	(0.0007–0.0336)	BS3 Strong	0.0095	(0.0024–0.0377)	BS3 Strong
Tejura: SGE FDR + directionality	Abnormal	depleted	ClinVar 2025 (missense only)	13/13	1/11	11.00	(1.70–71.28)	PS3 Moderate	5.57	(1.56–19.91)	PS3 Moderate
Tejura: SGE FDR + directionality	Normal	unchanged	ClinVar 2025 (missense only)	0/13	10/11	0.0408*	(0.0027–0.6260)	BS3 Strong	0.0857	(0.0127–0.5763)	BS3 Moderate
GMM score cutoffs	Abnormal	score ≤ -0.0286	KURIS/NDD vs B/LB missense	14/17	0/28	46.72*	(2.97–736)	PS3 Strong	22.56	(3.24–157)	PS3 Strong
GMM score cutoffs	Normal	score ≥ -0.0197	KURIS/NDD vs B/LB missense	3/17	28/28	0.1765	(0.0632–0.4928)	BS3 Moderate	0.2302	(0.0967–0.5477)	BS3 Moderate
Waters: SGE FDR + directionality	Abnormal	depleted	KURIS/NDD vs B/LB missense	15/17	1/28	24.71	(3.58–171)	PS3 Strong	12.08	(3.12–46.76)	PS3 Moderate
Waters: SGE	Normal	unchanged +	KURIS/NDD	2/17	27/28	0.1220	(0.0331–0.4494)	BS3	0.1790	(0.0634–0.5053)	BS3

Assignment method	Functional class	Class definition	Control set	Path. (in/N)	Ben. (in/N)	LR	95% CI	Evidence	OddsPath	OddsPath 95% CI	OddsPath evid.
FDR + directionality		enriched	vs B/LB missense					Moderate			Moderate
Tejura: SGE FDR + directionality	Abnormal	depleted	KURIS/NDD vs B/LB missense	15/17	1/28	24.71	(3.58–171)	PS3 Strong	12.08	(3.12–46.76)	PS3 Moderate
Tejura: SGE FDR + directionality	Normal	unchanged	KURIS/NDD vs B/LB missense	2/17	23/28	0.1432	(0.0385–0.5325)	BS3 Moderate	0.2101	(0.0736–0.6003)	BS3 Moderate
Waters: SGE FDR + directionality	Abnormal	depleted	Waters ClinVar ≥2* (Table 1)	0/0	0/6	—	not computable (n=0)	—	—	—	—
Waters: SGE FDR + directionality	Normal	unchanged + enriched	Waters ClinVar ≥2* (Table 1)	0/0	6/6	—	not computable (n=0)	—	—	—	—
Waters: SGE FDR + directionality	Abnormal	depleted	Waters ClinVar ≥1* (Table 1)	7/7	0/6	13.12*	(0.9026–191)	PS3 Moderate	6.12	(0.9792–38.31)	PS3 Moderate
Waters: SGE FDR + directionality	Normal	unchanged + enriched	Waters ClinVar ≥1* (Table 1)	0/7	6/6	0.0673*	(0.0046–0.9932)	BS3 Moderate	0.1458	(0.0227–0.9351)	BS3 Moderate
Waters: SGE FDR + directionality	Abnormal	depleted	Waters Systematic (Table 1)	2419/2423	4/138	34.44	(13.11–90.47)	PS3 Strong	27.74	(11.73–65.60)	PS3 Strong
Waters: SGE FDR + directionality	Normal	unchanged + enriched	Waters Systematic (Table 1)	4/2423	134/138	0.0017	(0.0006–0.0045)	BS3 Very Strong	0.0021	(0.0009–0.0051)	BS3 Very Strong

Baseline: $LR = (a/N_{path})/(b/N_{ben})$; 95% CI is log-Wald. The OddsPath columns are a stability check and do not affect the baseline.

OddsPath = Tavtigian/Brnich likelihood ratio with +1 added to whichever in-class count is the error cell for that row -- benign (a false positive) for Abnormal rows: $((a/(n1+1))/((b+1)/(n2+1)))$; pathogenic (a false negative) for Normal rows: $((a+1)/(n1+1))/(b/(n2+1))$. Its 95% CI is log-Wald and 'OddsPath evid.' is its ACMG tier.

* Haldane–Anscombe correction (+0.5 to all four cells) applied when a functional-class cell is empty.